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Dublin City University

SEMESTER 1 EXAMINATIONS 2023/2024

MODULE: CA266 - Probability & Statistics

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing)
SHSAO	Study Abroad (Science & Health)

YEAR OF STUDY: 2,O

EXAMINER(S):

Dr. Graham Healy (Internal) (Ext:7364)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: **Answer 4 questions. All questions carry equal marks.**

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1 ***[TOTAL MARKS: 25]***

For each of these questions **explain your answer** i.e., show you reasoning.

Q 1(a) **[14 Marks]**

A computer system uses passwords that have five characters, and each character is one of 26 letters. You may assume that passwords are not case sensitive. Passwords are randomly chosen by the system e.g., "avxaa".

Determine the proportion of passwords that:

I. begin with a vowel i.e., a, e, i, o or u.
II. begin with a vowel **or** end with a consonant.
III. where the first, third, and last letters are the same.
IV. do not have any repeated letters.
V. consist entirely of the letters up to and including the letter "*j*" without any repetition.

Q 1(b) **[11 Marks]**

A batch of 50 components contains 10 that are defective. Three are selected at random **without replacement**.

I. What is the probability that the second component is defective given that the first was defective?
II. What is the probability that all three components are defective?
III. What is the probability that the first, second, or both components are defective?
IV. What is the probability that at least one of the three components are defective?

[End of Question 1]

QUESTION 2 **_[TOTAL MARKS: 25]_**

For each of these questions **explain your answer** i.e., show you reasoning.

Each day **X** cars arrive to a garage for repair.

The table below shows the number of cars that arrived at the garage over the past 365 days e.g., 1 car arrived per day for 134 of the days, 2 cars arrived per day for 67 of the days, etc.

Number of cars (**X**)	0	1	2	3	4	5	6
Number of days	134	134	67	22	6	1	1

If a car arrives to the garage, it has a 0.9 probability of being repaired. If it is not repaired on the same day it arrives, it is returned to the owner.

Q 2(a) **[10 Marks]**

What are the mean and variance of the expected number of cars that will arrive per day?

Q 2(b) **[10 Marks]**

Give values for the following probabilities:

i. P(fewer than 2 cars arrive in a day)

ii. P(3 cars are repaired | 3 cars arrived)

iii. P(4 cars arrive in a day | at least 1 car has already arrived that day)

iv. P(1 or more cars are repaired | 4 cars arrived)

Q 2(c) **[5 Marks]**

Assuming exactly **n** cars arrive to the garage each day, what is the expected variance of the number of cars that will not be repaired in a 365-day period?

[End of Question 2]

QUESTION 3 ***[TOTAL MARKS: 25]***

In a university department, 10% of people use Word, 30% use LaTeX, and 60% use HTML. From experience it is known that:

- 40% of the Word documents contain errors.
- 20% of the LaTeX documents contain errors.
- 20% of the HTML documents contain errors.

For each of these questions **explain your answer** i.e., show you reasoning.

Q 3(a) **[3 Marks]**
What is the probability that a document does **not** contain errors, given that it is written in Word?

Q 3(b) **[6 Marks]**
If a document is chosen at random, what is the probability that it will contain errors?

Q 3(c) **[9 Marks]**
A document is chosen at random and is found to contain errors. Is it most probable the document was written in Word, LaTeX, or HTML? **Explain your answer.**

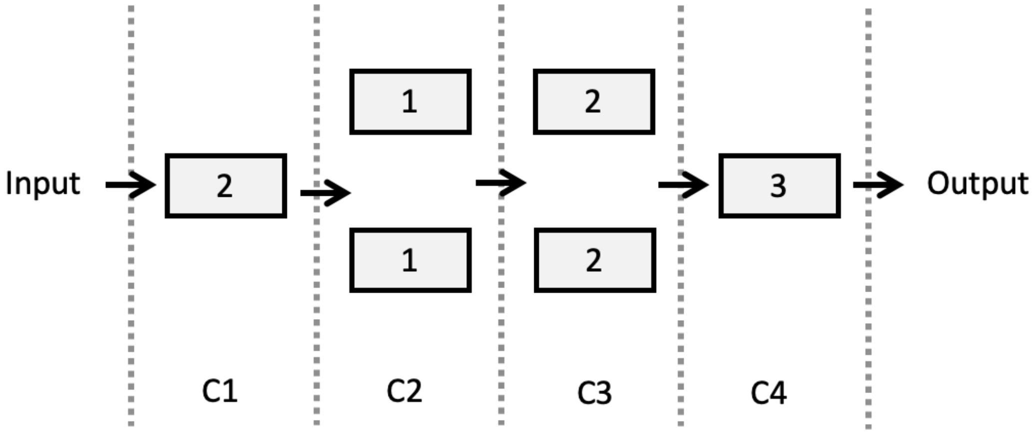
Q 3(d) **[7 Marks]**
A worker is inspecting documents, one by one, until they find a LaTeX or HTML document.

i. What's the probability that the first LaTeX or HTML document they find will be the 10th document they inspect?
ii. What's the probability that the first LaTeX or HTML document they find will be in the first 50 documents they inspect?

[End of Question 3]

QUESTION 4 ***[TOTAL MARKS: 25]***

The following system consists of a number of components in series and in parallel. The number in the box is the mean number of failures for that type of component in a **period of 1 year**. There are four (4) types of components in this system: C1, C2, C3 and C4. Assume the probability of the failures follow a Poisson distribution.



For each of the questions below **explain your answer** i.e., show your reasoning.

Q 4(a) **[4 Marks]**
For each component, give the probability that it will still function at the end of the 1-year period.

Q 4(b) **[4 Marks]**
For each component, give the probability that it will fail 3 times in a 1-year period.

Q 4(c) **[7 Marks]**
What is the probability that the whole system will still function at the end of the 1-year period?

Q 4(d) **[6 Marks]**
How long would component C1 need to be in operation for there to be a .9995 probability that a failure of this component will occur?

Q 4(e) **[4 Marks]**
An engineer is studying component C4. Given that the component has not failed in the past 3 months, what is the probability that it will fail in the following 3 months?

[End of Question 4]

QUESTION 5 *[TOTAL MARKS: 25]*

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each of these questions **explain your answer** i.e., show you reasoning.

Q 5(a) **[12.5 Marks]**

We throw two 6-faced dice, let the total score be X.

I. What are the probabilities for each value of X?
II. If we are told that the first throw was greater than 3, what is the probability that $X < 11$?
III. If we are told that **both** throws of the dice were an even number, what is the probability that $X > 6$?

Q 5(b) **[12.5 Marks]**

A well-shuffled deck of cards contains 52 cards, comprised of 4 suits: 13 diamonds, 13 spades, 13 clubs, and 13 hearts. Within each suit, each card is labelled Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and King.

I. If I draw 10 cards from the deck (without replacement), what is the probability that 3 of these will be hearts?
II. If I draw 10 cards from the deck (without replacement), what is the probability that 2 or more of these will be diamonds?
III. If I draw 10 cards from the deck (without replacement), what is the probability that at least one of the cards will be an Ace or King?

Q 5(c) **[12.5 Marks]**

Person A and person B have taken part in a competition every week for the past number of years. We are told that the average score for both contestants is 50. We are told the variance of the scores for Person A is 9. We are not told the variance of the scores for person B. You can assume the scores for both person A and person B are normally distributed.

I. What is the probability that person A scored above 56?
II. What is the probability that person B scored above 59, if we are told their score is above 41?
III. We are told there is a 0.9332 probability that person B scored below 53. What is the standard deviation for person B?

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

$${}^{n}C_k = \frac{{}^{n}P_k}{k!} = \frac{n!}{k!(n-k)!}$$	
$${}^{n}P_k = n(n-1)\ldots(n-k+1) = \frac{n!}{(n-k)!}$$	
$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$	
$$P(A\|B) = \frac{P(A)P(B\|A)}{P(B)}$$	$$P(A \cap B) = \begin{array}{l} P(A)P(B\|A) \\ P(B)P(A\|B) \end{array}$$
$$V(X) = \sum_x (x-\mu_x)^2 p(x)$$	$$E(X) = \sum_x xp(x)$$
$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1}\, p$
$$P(X=x) = \frac{\binom{M}{x}\binom{L}{n-x}}{\binom{N}{n}}$$	$$P(X=x) = e^{-\lambda}\frac{\lambda^x}{x!}$$

[END OF APPENDICES]

[END OF EXAM]